

Major Project (CSL8494)

Backend Developer internship at Eazotel



Presented by:

Group no.: 16

Name

Roll No.

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Under supervision of: **Divyanshu Sharma (Tech Lead at Eazotel) and Sachin Kapoor (CEO)**



INTRODUCTION

What is the vision of Eazotel?
Empowering hotels to operate independently with **minimal** reliance on OTAs for bookings and other SaaS for booking entries and other website for website creation, ensuring full control over their business operations and bringing all required operation at single place for easy handling

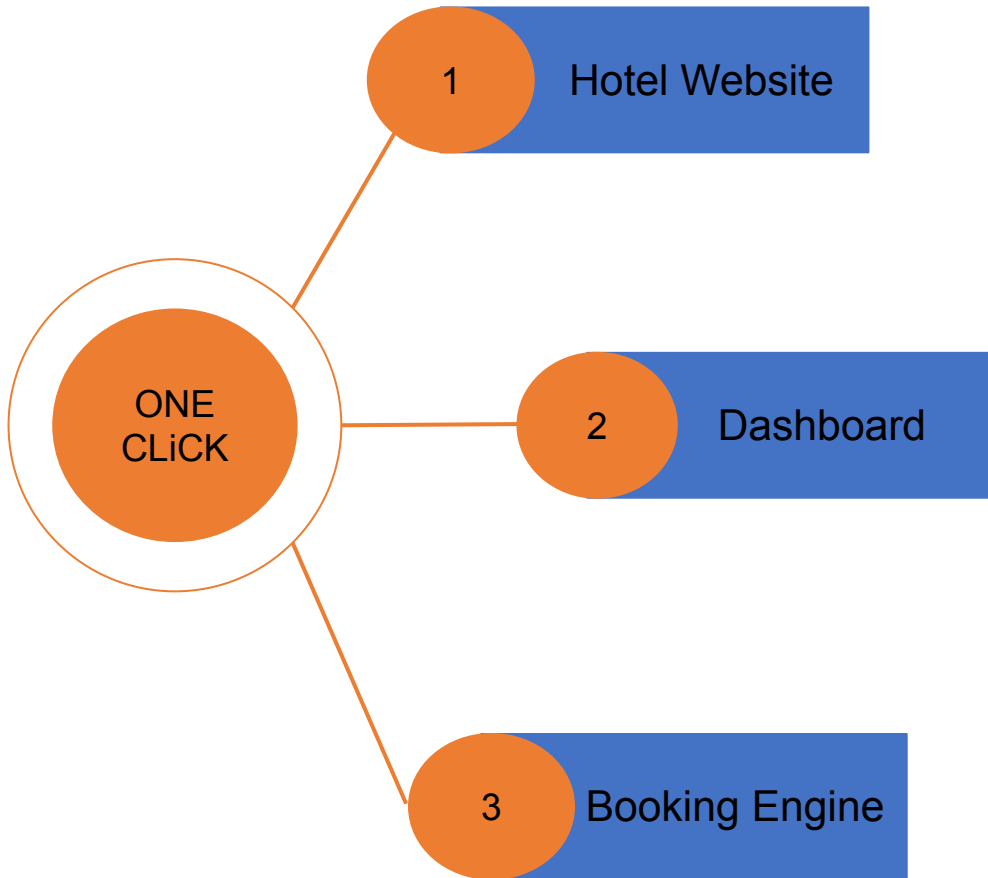


What is Eazotel:

Eazotel, is a software-as-a-service (SaaS) solution that provides an all-in-one hospitality management system designed to streamline operations and improve guest experiences for hotels, resorts, and other lodging establishments. Eazotel's mission is to revolutionize the hospitality industry by offering a comprehensive suite of features that simplify management, reduce costs, and improve guest satisfaction.



What is Eazotel (contd):



In today's fast-paced digital landscape, standing out online is crucial. That's where Eazotel comes in. With Eazotel, creating stunning websites, dashboards, and booking engines is as easy as 1-2-3 – all in less than a minute!

Imagine having the power to craft SEO-optimized, server-side rendered platforms that not only look great but also drive results. With Eazotel, you can do just that.

Say goodbye to long hours spent coding and hello to instant, professional-grade websites with Eazotel.

My MAJOR tasks:.

- 1. Eazotel website creation process implementation by utilizing BeautifulSoup for web scraping and other third party APIs for content in database**
- 2. Multi-location hotel handling process integrate with the system and also with the existing system.**
- 3. Room number allocation process algorithm implementation.**
- 4. Inventory management system implementation and integration with the dashboard.**
- 5. Implementation and integration with wati for whatsapp APIs.**
- 6. Implementation and integration of phonepe, razorpay with dashboard.**
- 7. Improvement of CMS(content management system) APIs and integration from dashboard.**
- 8. Implementation of spreadsheet APIs and integration with dashboard**
- 9. Integration of implementation of booking cancellation system with booking engine**



Example of room number allocation:

EAZOTEL
Spice Heaven
Wind 2 BHK,Mussoorie, 246761

View WebsiteView Booking Engine

Dashboard

Property Content

CMS

Booking Engine

Social Media

Reservation Desk

Front Desk

Channel Manager

SEO Manager

Payments

Whatsapp

Leads Management

User Management

Today's

Date RangeSingle Booking S

Booking Date From:
mm/dd/yyyy

Show Bookings

Total Bookings 1

Booking Id	Gues								
B2024021800001	Divyanshu Sharma	170	2024-03-04	2024-03-05	6844	SUCCESS	Check In		

Booking ID :B2024021800001

Order ID: NE2wdHDZVunnTeh6dzBTTR

Guest Name: Divyanshu Sharma

Email: test@gmail.com

Phone: +919119059286

Addr: Dhampur, India

Adults: 1

Kids: 0

Bookings: Delux 1

Payment Status: SUCCESS

Cancel Booking

Check In: 2024-03-04

Check Out: 2024-03-05

Checked In: Pending

Checked Out: Pending

Price:

Principle:	5800
Tax:	1044
Total:	6844
Paid:	6844

Example of room number allocation:

 **Spice Heaven**

Wind 2 BHK,Mussoorie, 246761

 [View Website](#) [View Booking Engine](#) 

 Dashboard

 Property Content

 CMS

 Booking Engine

 Social Media

 Reservation Desk

 **Front Desk**

 Channel Manager

 SEO Manager

 Payments

 Whatsapp

 Leads Management

 User Management

Zoom: ☒ Month ☐ Week Welcome! ☒ Auto width

DEMO

March 2024

Room Name	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Abhijeet test																														
101				Divyanshu Sh																										
102																														
103																														
104																														
105																														

Example of Wati(WhatsApp APIs)

EAZOTEL

Spice Heaven

Wind 2 BHK,Mussoorie, 246761

Q

View Website

View Booking Engine

Dashboard

Property Content

CMS

Booking Engine

Social Media

Reservation Desk

Front Desk

Channel Manager

SEO Manager

Payments

Whatsapp

Leads Management

User Management

+ Add new Contact

Templates Messages

918799934000

918799934000

919797005535

919797005535

abhaypts

919140823654

Fielmente Hospitality Marketing Agencysad

919501868775

ABC

919756143731

Unknown

919765133668

918799931000

918799931000

919119059286

919119059286

abhaypts

919140823654

...

I appreciate your interest in Eazotel a product by fielmente - a leading Hospitality Website Development Company based in Gurugram and London. We understand the importance of a strong online presence and offer comprehensive website development, booking engine, hosting, email service, a dashboard to manage room rates and bookings We appreciate you filling out our contact form. Please let us know your available time to schedule a call and take this journey ahead. Thank you!

✓2/27/2024, 1:49:11 AM

The chat has been initialized by contact abhaypts (919140823654)

2/27/2024, 7:11:18 AM

Working

2/27/2024, 7:11:18 AM

The ticket status has been set as Open by agent Bot

2/27/2024, 7:11:19 AM

The chat has been expired (after 24 hours of last received message)

2/28/2024, 7:20:34 AM

Send Message

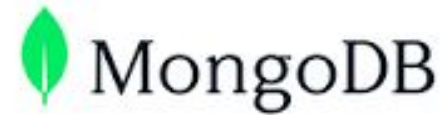
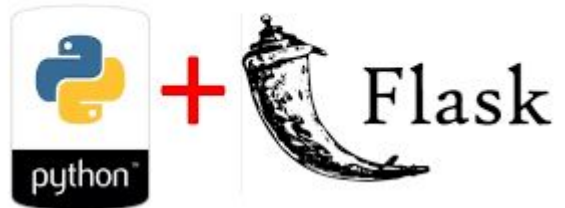
Technologies Used



Amazon EC2

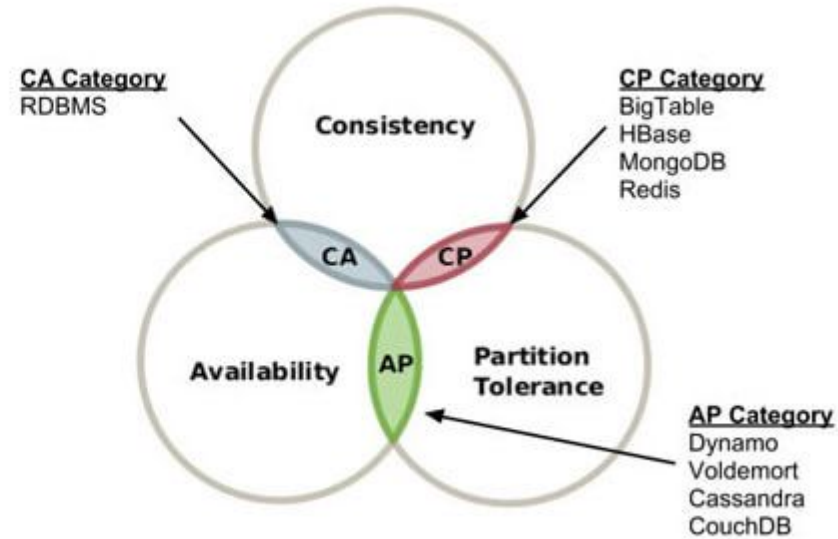


REST API using Flask



System Design

CAP Theorem



- MongoDB provides of CP databases.

System Design (contd.)

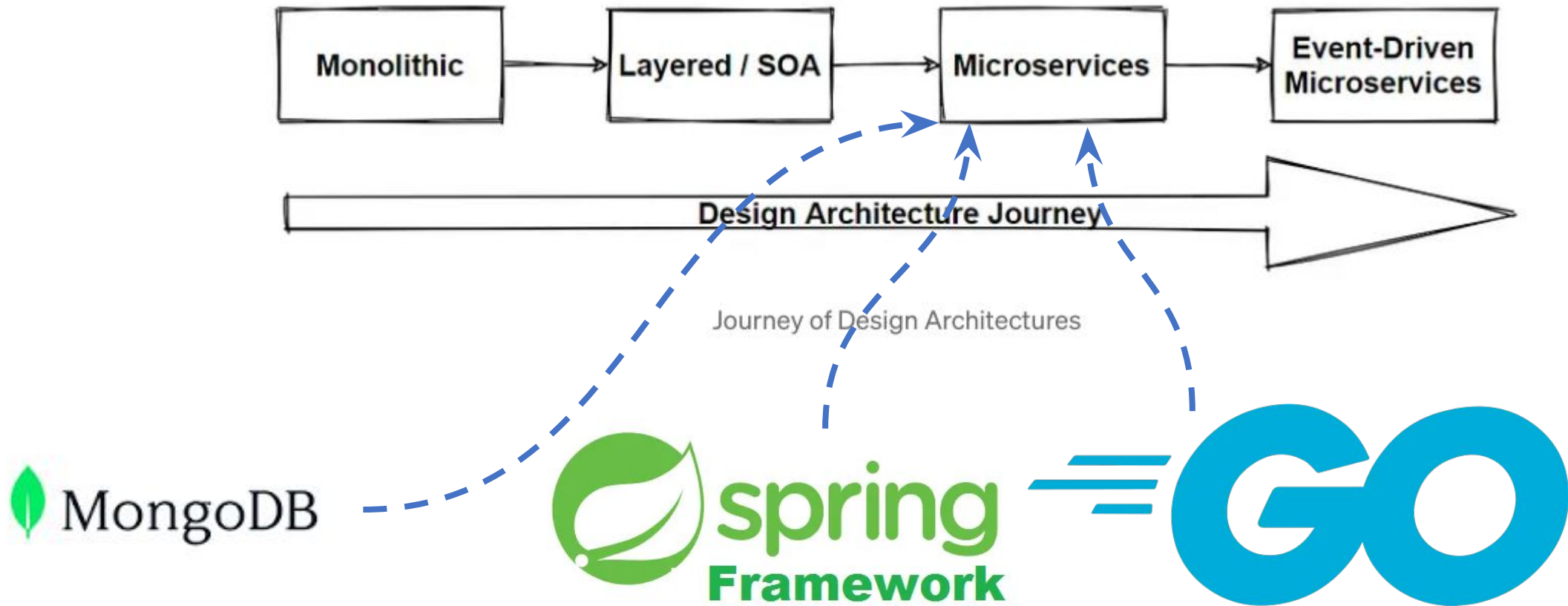
Summary of differences: monolithic vs. microservices

Category	Monolithic architecture	Microservices architecture
Design	Single code base with multiple interdependent functions.	Independent software components with autonomous functionality that communicate with each other using APIs.
Development	Requires less planning at the start, but gets increasingly complex to understand and maintain.	Requires more planning and infrastructure at the start, but gets easier to manage and maintain over time.
Deployment	Entire application deployed as a single entity.	Every microservice is an independent software entity that requires individual containerized deployment.
Debugging	Trace the code path in the same environment.	Requires advanced debugging tools to trace the data exchange between multiple microservices.
Modification	Small changes introduce greater risks as they impact the entire code base.	You can modify individual microservices without impacting the entire application.
Scale	You have to scale the entire application, even if only certain functional areas experience an increase in demand.	You can scale individual microservices as required, which saves overall scaling costs.

The system built is based on monolithic architecture

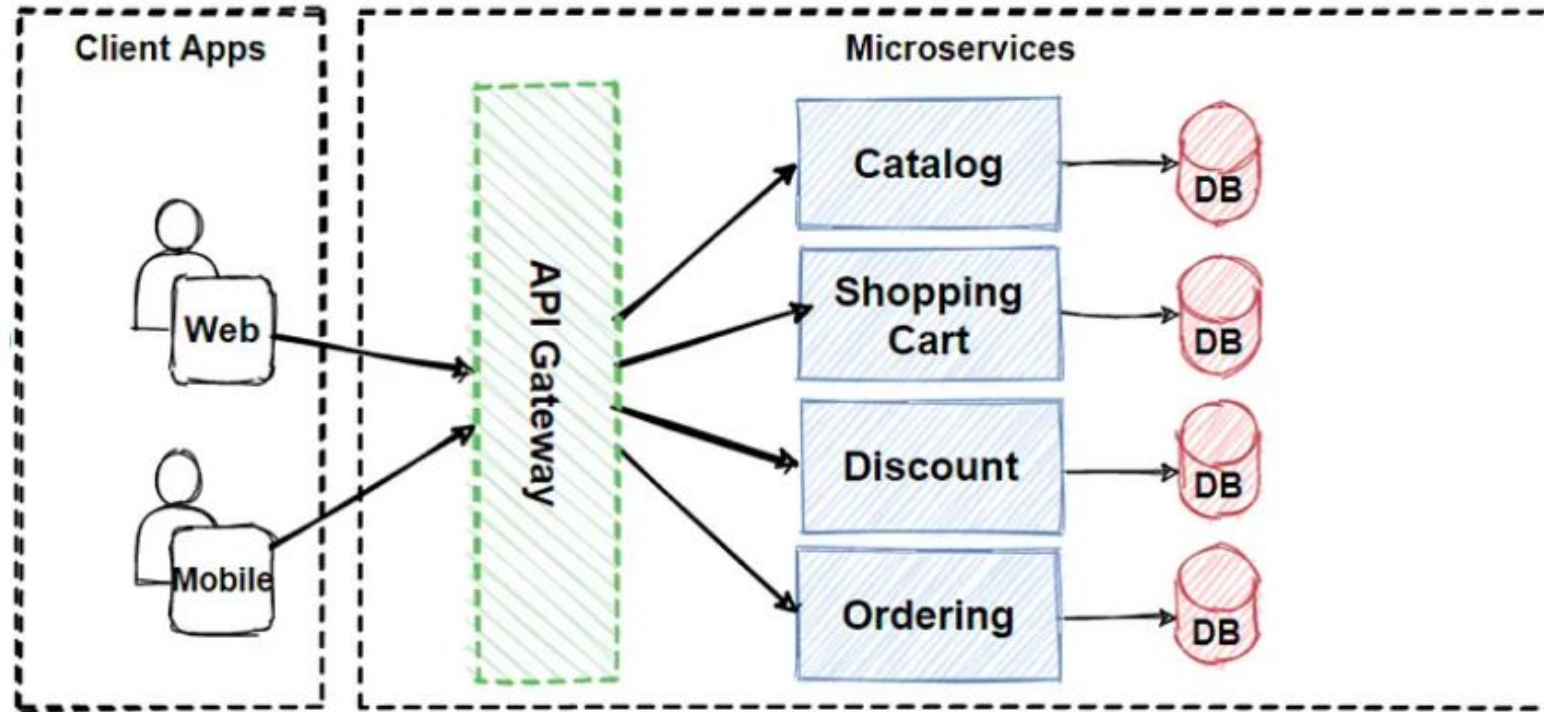


Future Improvements



Future Improvements (contd..)

Microservices Architecture - Api Gw



Example of monolithic architecture

Challenges Faced:

- Understanding pre-written code
- Understanding the whole dynamics of the system
- Code reintegration
- Dependencies and installation
- Debugging and troubleshooting



References:

- [1] Microservices architecture and framework understanding
- [2] MongoDb architecture and Documentation
- [3] Monolithic architecture shift to microservices architecture



THANK YOU

